

6(a)	${}^{14}_7\text{X}$	B1
	${}^0_{-1}\text{e}^-$	B1
6(b)(i)	$\text{d} \rightarrow \text{u} + \text{e}^- + \bar{\nu}$ or $\text{udd} \rightarrow \text{uud} + \text{e}^- + \bar{\nu}$	B1
6(b)(ii)	$-1/3 (e) = +2/3 (e) - 1(e) (+0)$ or $2/3 (e) - 1/3 (e) - 1/3 (e) = 2/3 (e) + 2/3 (e) - 1/3 (e) - 1(e) (+0)$	B1
6(c)(i)	electrons / β -particles (emitted from the nucleus) have a (continuous) range of / different (kinetic) energies	B1
6(c)(ii)	the (emitted) neutrinos take varying amounts of the (same total) energy (released in the decay)	B1